SPRING MID SEMESTER EXAMINATION-2020

Design and Analysis of Algorithm

CS-2012

Time :1:30 **Marks: 20**

Answer Question 1 and any three (2,3,4,5)

1 a. Compare the order of growth n! and 2!

b. Define Ω-notation

c. Give an non-recursive algorithm to find out the largest element in a list of n numbers.

d. What is the average case complexity of linear search algorithm?

e. Define fractional knapsack problem.

(Any three)

2. a) Find the upper bound of recurrences given below by substitution method. 2.5

$i)\ T(n)=2T(n/2)+n$

b) Explain Master theorem for decreasing function and dividing function. 2.5

3. a) Define heap. Explain the properties of heap. Using heap sort 20,35,18,8,14,41,3,39. 5

4. a) Consider 5 items along their respective weights and values I = (I1,I2,I3,I4,I5), w = (5, 10, 20, 30, 45) , v = (30, 20, 80, 90,150), The capacity of knapsack W = 70. Find the maximum profit. 2.5

b) Write an algorithm and derive the time complexity to maximize total profit for an array of jobs where every job has a deadline and associated profit if the job is finished before the deadline. It is also given that every job takes single unit of time, so the minimum possible deadline for any job is 1. How if only one job can be scheduled at a time. 2.5

5. a) Sort the given series of number using quick sort 29, 23, 17, 57, 34, 89, 65, 27 where first element is pivot element. What is the difference between quick sort and merge sort. 3

b) Write an algorithm for insertion sort. Derive the best case and worst case time complexity. 2